

CELEBAL TECHNOLOGIES - WEEK 1 ASSIGNMENT SUMMARY

Name : N.Varija Reddy

Email : varijareddy@gmail.com

Dataset details:

Link : <https://www.kaggle.com/datasets/anvitkumar/shopping-dataset>

File : Combined_dataset.csv

Original : 1000 rows x 24 columns

Final : 1000 rows x 26 columns

Steps involved:

Step 1 - Loaded Dataset

- > Loaded CSV using `pd.read_csv()`

- > 1000 rows and 24 columns loaded successfully

Step 2 - Data Exploration

- > Viewed `head()` and `tail()`

- > Checked shape, columns and data types

Step 3 – Handling Missing Values

- > Identified nulls using `isnull().sum()`

- > Numeric columns filled with median

- > Text columns filled with Unknown

- > Missing values remaining : 0

Step 4 - Basic Operations

- > Filtered products with `rating > 4`

- > Selected important columns for analysis

- > Columns selected : title, category, rating,
initial_price, discount, final_price, seller_name

Step 5 - Remove Duplicates

- > Checked duplicates using `duplicated().sum()`

-> Removed using drop_duplicates()

Step 6 - Derived Column

-> Added quantity column assumed = 1

-> Created total_amount = final_price x quantity

-> New column added successfully

Step 7 - Save Cleaned Dataset

-> Saved as shopping_cleaned.csv

-> Final shape : 1000 rows x 26 columns

Results:

Final Rows : 1000

Final Columns : 26

Missing Values : Filled with median and Unknown

Duplicates Removed : Yes

New Column Added : total_amount

Output File : shopping_cleaned.csv